

Week 2: From Concepts to Data

PLSC 309 Quantitative Political Analysis

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September 1, 2026

Problem Set 1

- **Problem Set 1** is now posted on Canvas.
- Due **next Tuesday at 11:59 p.m. (Sep 8th)**.
- There are **5 problem sets**, each worth **7 points**.
- Each problem set includes a **1-point bonus question**, so you may earn up to **8 points** on each problem set.
- The maximum total score across all five problem sets remains **35 points**.

APSA Conference Travel

- I will be traveling to the **APSA Conference** this week.
- **No office hours on Wednesday.**
- **No class on Thursday.**

Start Thinking About Your Final Project

You can start thinking about a possible:

- **Research Topic** — What area of politics (social sciences) interests you?
- **Research Question** — What specific question do you want to answer?
- **Theory** — What do you think explains the outcome, and why?
- **Hypothesis** — What relationship do you expect between your IV and DV?
- **Data Source** — Where might you find the data needed to test your hypothesis?

It does not need to be perfect or final.

The goal is simply to start developing an idea that we can refine over the next few weeks.

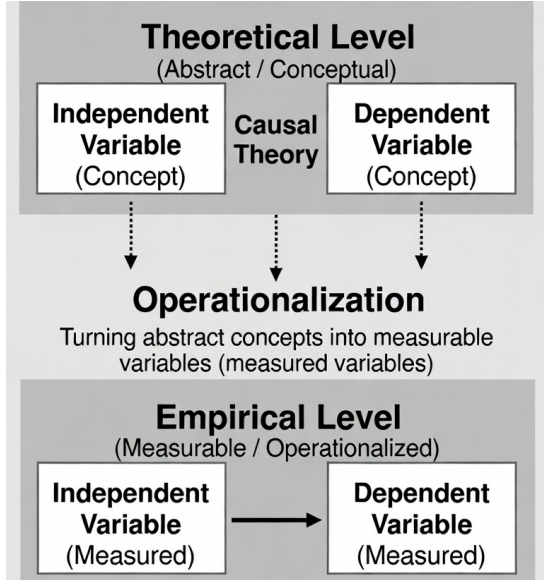
What Did We Learn Last Time?

- The Quantitative Research Process in Political Science

What Did We Learn Last Time?

- The Quantitative Research Process in Political Science
- Research Question
- Theory
- Hypothesis

From Theory (Concepts) to Empirics (Data)



From Theory (Concepts) to Empirics (Data)

- Concepts, Measures, Variables
- Variable Types
- Data Structures
- Data Sources

Part 1. Concepts, Measures, Variables

Concept

- A concept is an abstract idea or phenomenon that we want to describe or explain.

E.g., democracy

Conceptualization

- Conceptualization is the process of defining exactly what we mean by a concept.

E.g., what do we mean by democracy?

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Possible conceptualizations:

- *Minimalist*: Competitive elections.
- *Liberal*: Elections, civil liberties, and constraints on executive power.
- *Participatory*: Liberal democracy plus broad political participation.

Different researchers may conceptualize the same concept differently.

Measure

- A measure is a rule or procedure used to represent a concept empirically.

The same concept can have multiple measures.

Example 1: Freedom House

Measure

A country or territory is awarded 0 to 4 points for each of 10 political rights indicators and 15 civil liberties indicators, which take the form of questions; a score of 0 represents the smallest degree of freedom and 4 the greatest degree of freedom. (on a 0-100 scale)

[Freedom in the World 2026 Methodology Questions](#)

Example 2: Polity5

Measure

- *Autocracy Score (0–10)*: Rates the extent to which political power is concentrated in the executive, executive recruitment is closed or noncompetitive, and political competition is restricted.
- *Democracy Score (0–10)*: Rates the extent to which executive recruitment is competitive, political participation is open and competitive, and the executive is constrained by other political institutions.
- *Polity Score (-10 to +10)*: Calculated by subtracting the autocracy score from the democracy score:

$$\text{Polity Score} = \text{Democracy Score} - \text{Autocracy Score}$$

[Polity5 Users' Manual](#)

Example 3: V-Dem (Varieties of Democracy)

Measure

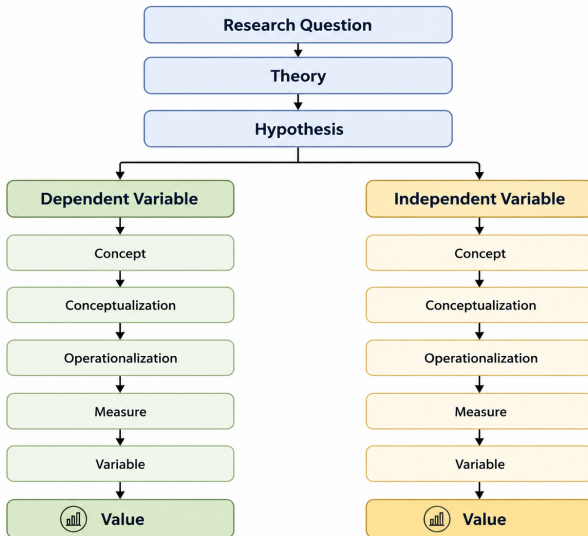
- *Expert-Coded Indicators*: More than 4,000 country experts evaluate over 500 indicators of political institutions, rights, participation, deliberation, and equality.
- *Multidimensional Measurement*: V-Dem treats democracy as consisting of multiple distinct dimensions rather than a single continuum.

Index	Variable	Scale
Electoral Democracy	v2x_polyarchy	0–1
Liberal Democracy	v2x_libdem	0–1
Participatory Democracy	v2x_partipdem	0–1
Deliberative Democracy	v2x_delibdem	0–1
Egalitarian Democracy	v2x_egaldem	0–1

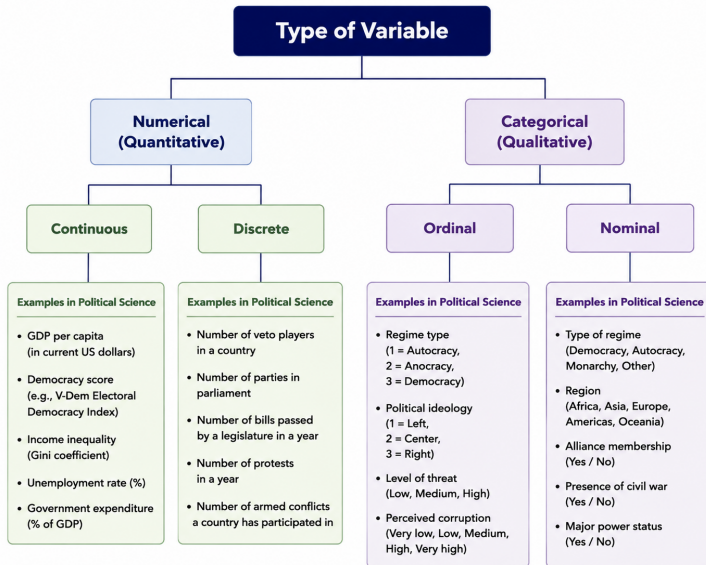
Variable - A variable is a measurable characteristic whose values differ across observations.

	Example
Concept	Democracy
Measure	V-Dem Electoral Democracy Index
Variable (Name)	Electoral Democracy Score (v2x_polyarchy)
Value	0.82 (United States, 2024)

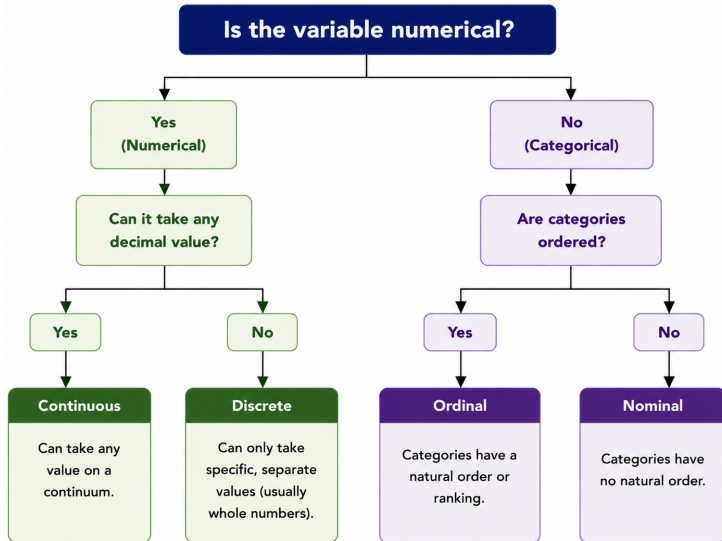
Translating Abstract Concepts into Measurable Variables



Part 2. Types of Variables



Variable Types (Decision Tree)



In-Class Exercise 1

Variable	Type
1. Country of residence	_____
2. Political party identification (Democrat, Republican, Independent)	_____
3. Ideology (Very Liberal, Liberal, Moderate, Conservative, Very Conservative)	_____
4. Annual household income (US dollars)	_____
5. Number of children	_____

In-Class Exercise 1 (Answer)

Variable	Type
1. Country of residence	Nominal
2. Political party identification (Democrat, Republican, Independent)	Nominal
3. Ideology (Very Liberal, Liberal, Moderate, Conservative, Very Conservative)	Ordinal
4. Annual household income (US dollars)	Continuous
5. Number of children	Discrete

In-Class Exercise 2

Variable	Type
6. Number of protests in a country last year	_____
7. Democracy score (V-Dem Electoral Democracy Index)	_____
8. Regime type (Democracy, Electoral Autocracy, Closed Autocracy)	_____
9. Satisfaction with democracy (Very dissatisfied → Very satisfied)	_____
10. GDP per capita	_____

In-Class Exercise 2 (Answer)

Variable	Type
6. Number of protests in a country last year	Discrete
7. Democracy score (V-Dem Electoral Democracy Index)	Continuous
8. Regime type (Democracy, Electoral Autocracy, Closed Autocracy)	Ordinal
9. Satisfaction with democracy (Very dissatisfied → Very satisfied)	Ordinal
10. GDP per capita	Continuous

Part 3. Types of Data

From Variables to Data

Once we have **variables**, we organize them into a **dataset**.

- A dataset is simply a collection of data (variables organized across observations).

Country	Year	Democracy	GDP per Capita	Population (Millions)
United States	2024	0.82	85,000	340
Canada	2024	0.91	62,000	40
Germany	2024	0.89	58,000	84
Brazil	2024	0.71	11,000	216
China	2024	0.08	13,000	1,410

- Rows = Observations
- Columns = Variables
- Cells = Values

Types of Data: Cross-Sectional vs. Time-Series

Cross-Sectional Data

Multiple units observed at one point in time

S1



Eleven

Mike

Dustin

Will

Lucas

Time-Series Data

One unit observed repeatedly over time

S1

S2

S3

S4

S5

Eleven



Types of Data: Panel Data

Panel Data

Multiple units observed repeatedly over time



Cross-Sectional Data

Character	Season	Age	Telekinetic Ability	Member of Party
Eleven	1	12	Yes	Yes
Mike	1	12	No	Yes
Dustin	1	12	No	Yes
Will	1	12	No	Yes
Lucas	1	12	No	Yes

Number of Observations = 5

Time-Series Data

Character	Season	Age	Telekinetic Ability	Member of Party
Eleven	1	12	Yes	Yes
Eleven	2	13	Yes	Yes
Eleven	3	14	Yes	Yes
Eleven	4	15	Yes	Yes
Eleven	5	16	Yes	Yes

Number of Observations =5

Panel Data

Character	Season	Age	Telekinetic Ability	Member of Party
Eleven	1	12	Yes	Yes
Eleven	2	13	Yes	Yes
Eleven	3	14	Yes	Yes
Eleven	4	15	Yes	Yes
Eleven	5	16	Yes	Yes
Mike	1	12	No	Yes
Mike	2	13	No	Yes
⋮	⋮	⋮	⋮	⋮
Lucas	3	14	No	Yes
Lucas	4	15	No	Yes
Lucas	5	16	No	Yes

Number of Observations = 5 units *times* 5 periods = 25

Pooled Cross-Sectional Data

Pooled cross-sectional data combine multiple independent cross-sectional samples collected at different points in time.

Respondent ID	Survey Year	Age	Ideology	Vote Intention
A001	2020	24	Liberal	Democratic
A002	2020	51	Conservative	Republican
A003	2020	37	Moderate	Democratic
B001	2024	21	Liberal	Democratic
B002	2024	45	Moderate	Republican
B003	2024	63	Conservative	Republican

Summary of Common Data Structures

Data Type	Units	Time Periods	Same Units Repeated?
Cross-sectional	Many	One	No
Time-series	One	Many	Yes
Panel	Many	Many	Yes
Pooled	Many	Many	No
Cross-sectional			

Unit of Analysis vs. Unit of Observation

	Unit of Analysis	Unit of Observation
Definition	The entity about which the researcher wants to draw conclusions.	The entity on which data are collected and recorded.
Determined by	Research question	Dataset

Part 4. Data Sources

1. Government Data (U.S.)

Governments collect large amounts of administrative and statistical data.

Examples

- **U.S. Census Bureau** → population, income, education, demographics
- **Bureau of Labor Statistics (BLS)** → employment, unemployment, wages
- **Bureau of Economic Analysis (BEA)** → GDP, personal income, economic growth
- **Federal Election Commission (FEC)** → campaign contributions and spending
- **State Election Agencies** → voter registration and election results
- **FBI Crime Data Explorer** → crime and law enforcement
- **Centers for Disease Control and Prevention (CDC)** → mortality and public health
- **Department of Education** → schools, enrollment, graduation, education outcomes
- **Congress.gov** → bills, legislation, and congressional activity
- **Data.gov** → datasets from U.S. federal government agencies

2. Public Opinion Surveys

Surveys are especially useful when your variables involve:

- Political attitudes
- Ideology
- Vote choice
- Political knowledge
- Trust
- Policy preferences

Major Sources

- American National Election Studies (ANES)
- General Social Survey (GSS)
- Cooperative Election Study (CES)
- Pew Research Center
- World Values Survey (WVS)

3. Cross-National Data (International Organizations)

International organizations collect and harmonize data across countries.

Examples

- **World Bank** → GDP, population, development, governance
- **United Nations (UN)** → population, migration, development, environment
- **International Monetary Fund (IMF)** → inflation, government finance, exchange rates
- **World Trade Organization (WTO)** → international trade and tariffs
- **OECD** → economy, education, taxation, social policy
- **International Labour Organization (ILO)** → employment, wages, labor conditions
- **World Health Organization (WHO)** → health, mortality, health systems
- **UNHCR** → refugees, asylum seekers, forced displacement
- **UNESCO** → education, science, and culture

3. Cross-National Data (Specialized Data Sources)

Many research organizations and academic projects collect data on specific political phenomena.

Democracy & Political Institutions

- **V-Dem** → democracy, elections, institutions, civil liberties
- **Freedom House** → political rights and civil liberties
- **Polity** → regime characteristics and political institutions

Conflict & International Relations

- **UCDP** → armed conflict, civil war, conflict fatalities
- **Correlates of War (COW)** → wars, disputes, alliances, national capabilities
- **SIPRI** → military spending, arms transfers, peace and security

Public Opinion

- **World Values Survey (WVS)** → values, political attitudes, social attitudes
- **Afrobarometer** → public opinion in Africa
- **Asian Barometer** → public opinion in Asia
- **Latinobarómetro** → public opinion in Latin America
- **European Social Survey (ESS)** → attitudes and behavior in Europe

4. Data Repositories

Search a data repository.

ICPSR - Large archive of social science datasets

Harvard Dataverse - Datasets shared by researchers

Replication Data - Data and code accompanying published research

5. Build Your Own Data

Sometimes the data you need do not already exist.

You can collect and construct your own dataset from:

- **Websites** → government records, organizations, news, online platforms
- **Digital Archives** → historical documents, newspapers, speeches
- **Social Media** → X/Twitter, Facebook, Reddit, YouTube
- **News Media** → articles, headlines, coverage of political events
- **Political Documents** → laws, bills, speeches, party manifestos
- **APIs** → structured data provided directly by online platforms
- **Web Scraping** → automatically extracting information from webpages

Is This Dataset Right for My Project?

Before using a dataset, check:

① Unit of Observation

- What does each row represent?

② Variables

- Does it contain my IV and DV?

③ Coverage

- Does it cover the population, places, and time period I want to study?

④ Accessibility

- Can I download it?

⑤ Documentation

- Is there a codebook explaining the variables?

Key Takeaways

Types of Variables

- Variables can be **numerical** or **categorical**.
- Numerical variables can be **continuous** or **discrete**.
- Categorical variables can be **nominal** or **ordinal**.

Types of Data

- Common data structures include **cross-sectional**, **time-series**, **panel**, and **pooled cross-sectional** data.
- The **unit of analysis** is what we want to draw conclusions about; the **unit of observation** is what each observation in the dataset represents.